

CN Unit 5 — Application Layer & Security

Complete Notes

Computer Networks (CN) · Unit 5 · Application Layer & Security · 2024

Overview

These notes cover **Application Layer & Security**. From bits on the wire to HTTP — layered network architectures, routing, transport protocols and the applications on top. Each section states the core idea, a worked intuition and the points examiners look for.

1. DNS

DNS resolves domain names via a hierarchy of root, TLD and authoritative servers.

Why it matters: DNS appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of dns in one line.
- A small example showing dns in practice.
- Common mistakes students make and how to avoid them.

2. HTTP

HTTP is a stateless request-response protocol; persistent connections and caching improve performance.

Why it matters: HTTP appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of http in one line.
- A small example showing http in practice.
- Common mistakes students make and how to avoid them.

3. Email Protocols

SMTP sends mail while POP3 and IMAP retrieve it.

Why it matters: Email Protocols appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of email protocols in one line.
- A small example showing email protocols in practice.
- Common mistakes students make and how to avoid them.

4. Symmetric and Public Key Cryptography

DES and AES use shared keys; RSA uses public/private key pairs.

Why it matters: Symmetric and Public Key Cryptography appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of symmetric and public key cryptography in one line.
- A small example showing symmetric and public key cryptography in practice.
- Common mistakes students make and how to avoid them.

5. Digital Signatures

Digital signatures provide authentication, integrity and non-repudiation.

Why it matters: Digital Signatures appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of digital signatures in one line.
- A small example showing digital signatures in practice.
- Common mistakes students make and how to avoid them.

6. Firewalls

Packet-filtering and application-level firewalls enforce network security policies.

Why it matters: Firewalls appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of firewalls in one line.
- A small example showing firewalls in practice.
- Common mistakes students make and how to avoid them.

Quick revision

DNS · HTTP · Email Protocols · Symmetric and Public Key Cryptography · Digital Signatures · Firewalls